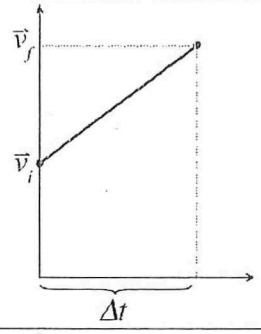
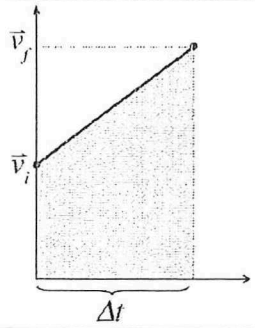
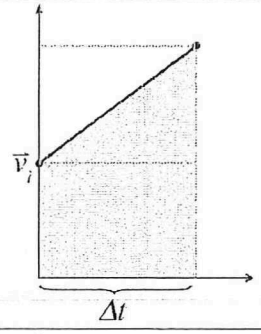
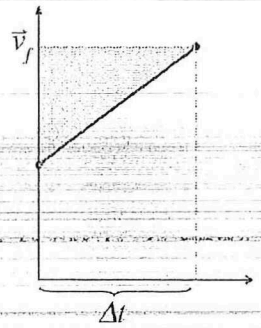
Kinematic Equations

The relevant quantities in the kinematic equations are: _____, _____, _____, _____, _____

The kinematic equations are five different ways to relate these quantities.

VT Graph	Derivation	Final form
	Solve for $\vec{a}$ using only variables from the graph. Rearrange your equation to isolate $\vec{v_f}$. Missing variable: _____	
	Solve for $\overrightarrow{\Delta d}$ using only variables from the graph. Hint: What is the area of the shaded trapezoid? Missing variable: _____	
	Solve for $\overrightarrow{\Delta d}$ without using $\vec{v_f}$. Hint: Find the area of the bottom rectangle and add the area of the above triangle. Missing variable: _____	
	Solve for $\overrightarrow{\Delta d}$ without using $\vec{v_i}$. Hint: Find the area of the entire rectangle and subtract the area of the shaded triangle. Missing variable: _____	
Solve for $\overrightarrow{\Delta d}$ without using Δt. Hint: Use the results from above and substitute. No graph is necessary. Missing variable: _____		

SPH3U: The BIG Five

A: The BIG Five – Revealed!

Here are the BIG five equations for uniformly accelerated motion.

The **BIG** Five	$\vec{v}_i$	$\vec{v}_f$	$\vec{\Delta d}$	$\vec{a}$	Δt
$\vec{v}_f = \vec{v}_i + \vec{a}\Delta t$					
$\vec{\Delta d} = \vec{v}_i\Delta t + \frac{1}{2}\vec{a}\Delta t^2$					
$\vec{\Delta d} = \vec{v}_f\Delta t - \frac{1}{2}\vec{a}\Delta t^2$					
$\vec{\Delta d} = \frac{1}{2}(\vec{v}_i + \vec{v}_f)\Delta t$					
$\vec{v}_f{}^2 = \vec{v}_i^2 + 2\vec{a}\vec{\Delta d}$					

1. **Observe.** Fill in the chart with √ and × indicating whether or not a kinematic quantity is found in that equation.

2. **Find a Pattern.** How many quantities are related in each equation?

3. **Reason.** If you wanted to use the first equation to calculate the acceleration, how many other quantities would you need to know?

4. **Describe.** Define carefully each of the kinematic quantities in the chart below.

$\vec{v}_i$	
$\vec{v}_f$	
$\vec{\Delta d}$	
$\vec{a}$	
Δt	

5. **Reason.** What condition must hold true (mentioned in the previous investigation) for these equations to give reasonable or realistic results?

B: As Easy as 3-4-5

Solving a problem involving uniformly accelerated motion is as easy as 3-4-5. As soon as you know **three** quantities, you can always find a **fourth** using a BIG **five**! Write your solutions carefully using our solution process. Use the chart to help you choose a BIG five. Here are some sample problems that we will use the BIG five to help solve. Note that we are focusing on certain steps in our work here – in your homework, make sure you complete all the steps!

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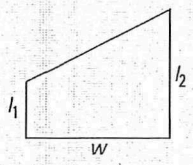
2.3 Check and Reflect

Key Concept Review

1. (a) If an object starts from rest, what is $\vec{v}_i$?
   (b) If an object comes to a stop, what is $\vec{v}_f$?
   (c) If an object experiences uniform motion, what is $\vec{a}$?

2. Figure 2.28 on page 48 shows the velocity-time graph of an object undergoing uniformly accelerated motion.
   (a) Use the units on the graph to show that the slope represents acceleration.
   (b) Show how the slope of the graph can be used to derive the first kinematics equation.

3. To calculate the displacement (area) from a velocity-time graph, you can use the formula for the area of a trapezoid, $A = \frac{1}{2}(l_2 + l_1)w$, which is the average of the parallel sides multiplied by the base. Show how this formula and the second kinematics equation are related.



Question 3

4. You can calculate the area under a velocity-time graph by considering that the total area under the graph is made up of a triangle and a rectangle.
   (a) Which part of the third kinematics equation represents the rectangle? Why?
   (b) Which part represents the triangle? Why?

Connect Your Understanding

5. How far will a humanoid robot travel in 3.0 s, accelerating at 1.0 cm/s² [forward], if its initial velocity is 5.0 cm/s [forward]?

6. What is the displacement of a logging truck accelerating from 10 m/s [right] to 20 m/s [right] in 5.0 s?

7. How far will a car travel if it starts from rest and experiences an acceleration of magnitude 3.75 m/s² [forward] for 5.65 s?

8. Determine the acceleration of a bullet starting from rest and leaving the muzzle 2.75×10^{-3} s later with a velocity of 460 m/s [forward].

9. An aircraft starts from rest and accelerates at 42.5 m/s² until it takes off at the end of the runway. How long will it take the aircraft to travel down the 2.6 km runway?

10. If a cyclist traveling at 14.0 m/s skids to a stop in 5.60 s, determine the skidding distance. Assume uniform acceleration.

11. Approaching a flashing pedestrian activated traffic light, a driver must slow down to a speed of 30 km/h. If the crosswalk is 150 m away and the vehicle's initial speed is 50 km/h, what must be the magnitude of the car's acceleration to reach this speed limit?

12. A train's stopping distance, even when full emergency brakes are engaged, is 1.3 km. If the train was travelling at an initial velocity of 90 km/h [forward], determine its acceleration under full emergency braking.

13. A rocket starts from rest and accelerates uniformly for 2.00 s over a displacement of 150 m [W]. Determine the rocket's acceleration.

14. A jet starting from rest reaches a speed of 241 km/h on 96.0 m of runway. Determine the magnitude of the jet's acceleration.

15. What is a motorcycle's acceleration if it starts from rest and travels 350.0 m [S] in 14.1 s?

16. Determine the magnitude of a car's acceleration if its stopping distance is 39.0 m for an initial speed of 97.0 km/h.

17. A typical person can tolerate an acceleration of about −49 m/s² [forward]. If you are in a car travelling at 110 km/h and have a collision with a solid immovable object, over what minimum distance must you stop so as to not exceed this acceleration?

18. Determine a submarine's acceleration if its initial velocity is 9.0 m/s [N] and it travels 1.54 km [N] in 2.0 min.

Reflection

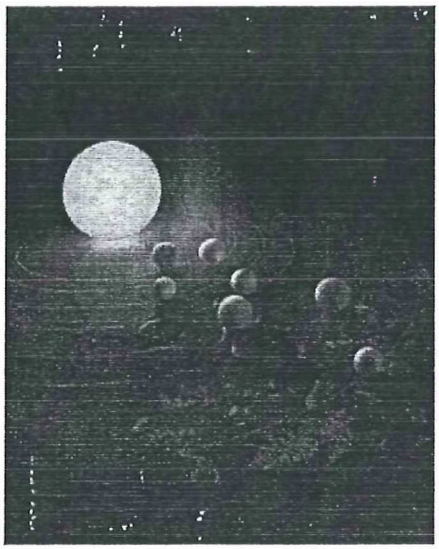
19. Why do you think it is important to understand the kinematics equations?

For more questions, go to PHYSICS•SOURCE

©P

Some Kinematics Examples

NASA announced the discovery of the TRAPPIST-1 system on Feb. 21, 2017. It consists of an ultra-cool dwarf star with 7 planets that are about the size of the earth! Three of them are in the habitable zones of their solar system.



The TRAPPIST-1 star, an ultra-cool dwarf, has seven Earth-size planets orbiting it. This artist's concept appeared on the cover of the journal Nature on Feb. 23, 2017.
Credits: NASA/JPL-Caltech

1. The system is 40.0 lightyears from ours. The fastest human made rocket can reach a speed of 70.0 km/s. How long would it take this rocket to reach TRAPPIST-1 if it travelled at top speed the entire trip? (TIP: There are 9.461×10^{15} meters in one light year.)

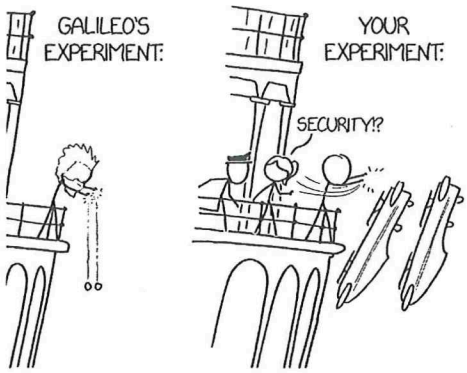
2. For a rocket to escape Earth's gravity, it must travel at 11.2 km/s.
    a. If a rocket started from rest, what must be its rate of acceleration to reach this speed by the time it gets to the stratosphere, 30.0 km above the Earth's surface?

    b. How long does this rocket take to reach a height of 10.0 km?

    c. How long does it take to travel from a height of 20.0 km to 30 km?

Acceleration due to Gravity

… or vertical motion… or 1d projectile motion



All things fall at the same rate. Galileo famously showed this by dropping two cannonballs of different masses off the leaning tower of Pisa. Both balls hit the ground at the same time.

On Earth lighter things appear to fall slower due to the effects of our atmosphere, mainly air resistance. Consider a feather and a hammer. On Earth the feather takes a lot longer to hit the ground. However, astronauts conducted the experiment on the moon (where there is no atmosphere) and were able to arrive at the same result as Galileo.
See this video: https://www.youtube.com/watch?v=ZVfhztmK9zI

If we were able to remove all the air in a certain room, the experiment would also work on Earth.
See this video: https://www.youtube.com/watch?v=E43-CfukEgs

So how fast do things fall? The accepted value to three significant digits is: $\mathbf{g} = 9.81\ m/s^2$ [down]

Important points:

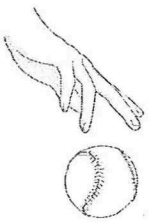
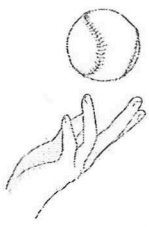
- This acceleration affects all objects.
- This value only applies when there are no other forces (such as air resistance) acting on the object.
- This value only applies near the surface of the Earth. Objects such as satellites and meteors would experience a different acceleration due to their distance from the Earth.
- Every planet, dwarf planet, or moon would have a different value of g that depends on their mass.

Very important point:

- This value is an acceleration, not velocity!
  An object's velocity does not change this value of g.

Example 1

What is the acceleration of each ball after it leaves the hand?

A ball falls off someone's hand.	A ball is thrown down.	A ball is thrown up.
		

Example 2

A light bulb falls from the ceiling. How long does it take to reach the ground? The ceiling is _____ from the ground.

Example 3

Romeo stands directly under Juliet's balcony and tosses a rose directly up at her at a speed of 10.0 m/s [up]. Juliet is 3.50 m above Romeo. Ignore air resistance.

a) At what time after Romeo tosses the rose should Juliet reach out to catch it?

b) What is the maximum height of the rose?

97 99 100

Note: Particularly challenging = #7, 11 + 12

Gravity Questions

1. A rock is dropped from the top of a tower and strikes the ground 4.0 s later.
   a) With what velocity did the rock hit the ground?
   b) What is the height of the tower?

2. An anti-aircraft shell is fired vertically upwards, from the ground, with a speed of 500 m/s (3 s.d.)
   a) How long does it take for the shell reach its maximum height?
   b) At what time(s) will the shell be 1.000 km above the Earth?

3. A ball is thrown upwards with a speed of 20.0 m/s from a roof that is 10 m (2 s.d.) above the ground.
   a) How much time will pass before the ball is travelling at a speed of 20.0 m/s, again?
   b) What is the total flight time of the ball?
   c) What is the velocity of the ball, as it hits the ground?

4. A rock is thrown straight upwards, from the top of a 120 m tall building. The rock hits the ground 5.9 s later.
   a) What was the velocity of the rock when it was first thrown?
   b) With what velocity did the rock hit the ground?

5. A stone is thrown vertically downwards, from a bridge, with an initial speed of 11.5 m/s. The rock strikes the water 3.2 s later.
   a) What is the velocity of the rock, just as it hits the water?
   b) What is the height of the bridge (above the water)?

6. An object is dropped from a very large height. How much farther does the object fall in 10.0 s, than in 1.0 s?

7. A helicopter that is hovering above a forest fire, dumps a large bucket of water. How far does the water fall during the third second (after it was dumped)?
   (⇒ between $t = 2s$ and $t = 3s$)

8. A person standing on the top of a 45 m tall building throws a perogie upwards, with a speed of 15 m/s. The unhappy perogie clears the edge of the building on the way back down, and plummets to the ground at the bottom. When the perogie hits the ground, it sadly ejects its contents of cheese and potatoes.
   a) How long will it take for the perogie to hit the ground?
   b) With what velocity did the perogie hit the ground?
   c) How long will it take for the perogie to reach its maximum height?
   d) What is the maximum height that the perogie reaches, above the top of the building?

9. A toy gun shoots a plastic dart upwards, from the top of a building, with a speed of 45 km/h. The building has a height of 23.2 m.
   a) When will the dart be 28.5 m above the ground?
   b) When will the dart be 37.0 m above the ground?
   c) When will the dart be 16.3 m above the ground?

10. A person is standing at the base of a building. They throw a 0.12 kg rock, straight up, with a speed of 28 m/s. The rock reaches a height of 12 m above the top of the building, before falling back down and landing on the top of the building.
   a) What it the height of the building?
   b) What is the velocity of the rock, just before it lands on the roof?

11. A person is standing on a balcony, which is 34.5 m above the ground. A person on the ground, below the balcony throws a ball vertically upwards. The ball passes the person, travelling upwards at 10.1 m/s. When is/was the ball 15.4 m above the ground?
   ← Relative to when it passed the person on balcony

12. Michael wants to find out how far it is, from the top of the building to his apartment. He sends his friend Jessica up to the roof, and has her drop a perfectly innocent book, so that it passes by his window, which is 1.2 m tall. He finds that it takes 0.024 s for the book to pass from the top of the window to the bottom of the window. How far is it from the top of the building to the bottom of the window?

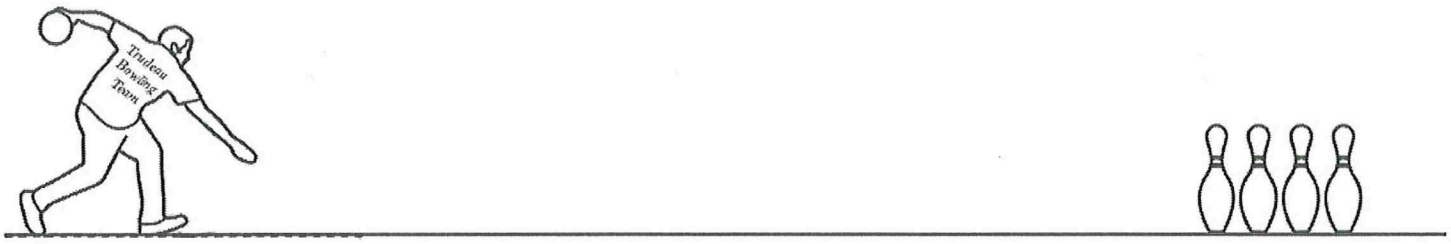
ANSWERS:

1. a) 39 m/s [down] (from 39.24 m/s [down]) b) 78 m (from 78.48 m)
2. a) 51.0 s (from 50.96... s) b) 2.04 s and 99.9 s (from 2.040... s and 99.895... s)
3. a) 4.1 s (from 4.0774... s) b) 4.5 s (from 4.5277... s)
   c) 24 m/s [down] (from 24.41... m/s [down])
4. a) 8.6 m/s [up] (from 8.6005... m/s [up]) b) 49 m/s [down] (from 49.278... s)
5. a) 43 m/s [down] (from 42.892 m/s [down]) b) 87 m (from 87.0272 m)
6. 490 m (from 485.595 m)
7. 24.5 m (from 24.525 m) note: times are assumed to be exact in this question, due to the wording
8. a) 4.9 s (from 4.92203... s) b) 33 m/s [down] (from 33.285... s) c) 1.5 s (from 1.529... s)
   d) 11 m (from 11.4678... s)
9. a) 0.54 s and 2.0 s (from 0.5372... s and 2.01115... s) b) never
   c) 3.0 s (from 3.0149... s)
10. a) 28 m (from 27.959... m) b) 15 m/s [down] (from 15.344... m/s [down])
11. 1.20 s before the ball passed the balcony AND 3.26 s after the ball passed the balcony
    (from 1.1961... s before the ball passed the balcony AND 3.25531... s after the ball passed the balcony)
12. 130 m (from 128.02170... m)

Kinematics Equations - Consecutive Intervals

The kinematics equations apply to situations in which the acceleration of an object does not change. If the acceleration changes over the course of an object's motion, then we have to break up the problem into multiple intervals and apply a separate set of equations for each interval.

Example



A bowler accelerates a bowling ball from rest at a rate of 4.78 m/s^2 for 2.00 m before releasing the ball. The ball then rolls at constant velocity for 18.29 m before it strikes the pins and comes to rest after 1.50 m. You may assume the ball accelerates at a constant rate while it is striking the pins.

a) What is the maximum velocity of the ball? [4.37 m/s [forward]]
b) How long does it take the ball to reach the pins after it is released? [4.18 s]
c) How long does it take the ball to come to rest after it strikes the pins? [0.172 s]

General Tips

- Sketch a diagram.
- Number each interval.
- Use subscripts to keep your variables organized.
- To avoid confusion with $\vec{v}_{initial}$ and $\vec{v}_{final}$, use $\vec{v}_1$, $\vec{v}_2$, $\vec{v}_3$, etc.
  - In the first interval let $\vec{v}_{initial} = \vec{v}_1$ and $\vec{v}_{final} = v_2$
  - In the second interval let $\vec{v}_{initial} = \vec{v}_2$ and $\vec{v}_{final} = v_3$
  - Continue for further intervals...
  - Can you explain why the above scheme would work?
- There must be a separate kinematics equation for each interval.

SPH3U PHYSICS (HW) #2, 3, 7, 8, 10

SPH3U PHYSICS
HOMEWORK – EQUATIONS OF MOTION

1. In an acceleration test for a sports car, two markers 0.30 km apart were set up along a road. The car passed the first marker with a velocity of 5.0 m/s [E] and passed the second marker with a velocity of 33.0 m/s [E]. Calculate the car's average acceleration between the markers. [1.78 m/s^2 [E])

2. A plane travels at 52 m/s [W] for 0.40 minutes down a runway. It then begins accelerating uniformly at 2.8 m/s^2 [W]. (accelerating for)
   a. What is the plane's velocity after 5.0 s? [66 m/s [W]]
   b. What is the total displacement of the plane for the entire time? (1543 m ≈ 1500 m [W])

3. A 100 m sprinter accelerates for the first 40 m reaching a top speed of 11.5 m/s and then runs at a constant speed for the remainder of the run. Find her acceleration during the first 40 m and the total time it takes for the 100 m. [1.7 m/s^2 [FWD], 12 sec) } Assume 2 S.D. minimum

4. A skier starting from rest accelerates uniformly downhill at 1.8 m/s^2 [forward]. How long will it take the skier to reach a point 95 m [forward] from the starting position? [10. s]

4 2. For a certain motorcycle, the magnitude of the braking acceleration is 39.2 m/s^2. If the bike is traveling at 32 m/s [S],
   a. how long does it take to stop? [0.82 s]
   b. how far does the bike travel during the stopping time? [13 m]

5. A car traveling down along a highway must uniformly reduce its velocity to 12 m/s [N] in 3.0 s. If the displacement traveled during that time interval is 58 m [N], what is the car's average acceleration? What is its initial velocity? [4.9 m/s^2 [S], 27 m/s [N]]

6. An arrow is accelerated for a displacement of 75 **cm** [forward] while it is on the bow. If the arrow leaves the bow at a velocity of 75 m/s [forward], what is its average acceleration while on the bow? [3750 m/s^2]

7. Jack and Jill ran down the hill. Both started from rest and accelerated steadily. Jack accelerated at 0.25 m/s^2 and Jill at 0.30 m/s^2. After running for 20.0 s, Jill fell down.
   a) How far did Jill get before she fell? [60 m]
   b) How far had Jack traveled when Jill fell? [50 m]
   c) How fast was Jack running when Jill fell? [5 m/s]
   d) How long was it after Jill fell that Jack ran into her and broke his crown (to the nearest second)? [1.9 s longer]

8. A car accelerates uniformly from rest at a rate of 2.0 m/s2 for 6.0 s. It then maintains a constant velocity for 0.50 min. Finally, the brakes are applied and the vehicle slows down at a uniform rate and comes to rest in 5.0 s. Find (a) the maximum velocity of the car and (b) the total displacement. [12 m/s [FWD], 430m [FWD]]

9. A skateboard travelling at 3.0 m/s comes to a hill and accelerates at 0.50 m/s/s for 5.0 s. It then coasts at a constant speed for 20.0 m and then hits a rough patch of the ground and comes to a stop in 3.5 s. What is the total displacement of the skateboard?

10. A turtle is moving with a constant acceleration along a straight ditch. He starts a stopwatch as he passes a fence post and notes that it takes him 10 s to reach a pine tree 10 m farther along a ditch. As the turtle passes the pine tree, his speed is 1.2m/s. How far was he from the fence post when he started from rest? [8 m]

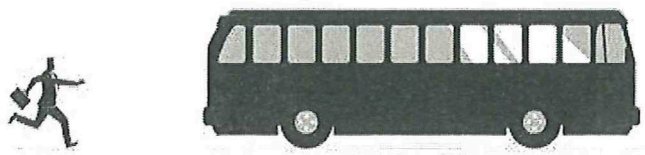
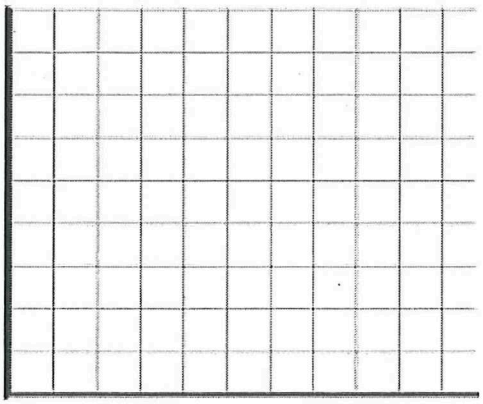
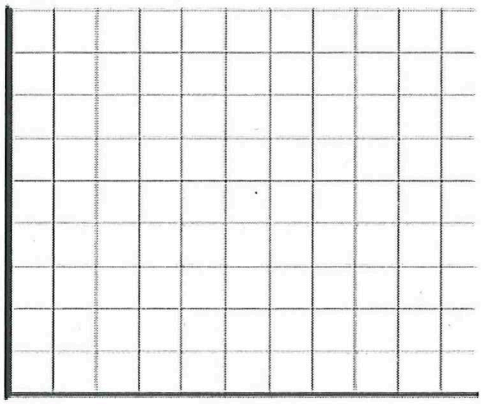
Kinematics Equations - Chase & Collision

These types of problems involve the motion of two objects.
Usually these problems involve the two objects meeting at a specific time or place in their motion.

- If the objects are moving in the same direction we refer to it as a "chase" situation.
- If the objects are moving toward each other we refer to it as a "collision".

Try this

How would the DT graphs look in each situation?

A person chases after a bus.	Two bumper cars headed toward each other.
	
	

You will usually have to solve for the position or time when the objects meet. Here are some guidelines to help you solve these types of problems.

How to set up chase and collision problems

1. Construct a line diagram to help you visualize the situation.
2. Choose a direction convention by defining a positive direction.
3. Let x be the position the objects meet.
4. Let y be the time that the objects meet.
5. Let the starting point of object 1 be the reference point.
6. Organize the variables using a T-chart.
7. Solve for x and y by substitution.
8. If a quadratic equation is used to solve for y, be sure to find both solutions and decide which one is more reasonable.

Here's a sample of how a T-chart might look like to organize your variables.

Object 1	Object 2
$\Delta d_1 = d_{1f} - d_{1i}$ $= x - 0$ $= x$	$\Delta d_2 = d_{2f} - d_{2i}$ $= x - d_{2i}$
$\Delta t_1 = y$	$\Delta t_2 = y$
List other known variables.	List other known variables.
Write down Kin. Eq.	Write down Kin. Eq.

Notice that your choice of Object 1 or Object 2 is arbitrary. Choose the one that makes the problem easier to solve.

In a chase situation, both Δd_1 and Δd_2 will have the same sign (positive).
In a collision situation, Δd_1 and Δd_2 will have opposite signs (one positive, one negative).
If your positions are properly defined, the equations will naturally give you the correct signs for each Δd.

Example 1
The Centennial Bridge connects New Brunswick to PEI. It spans a distance of 1,355 m. A car at one end of the bridge starts at rest and accelerates North at a rate of 1.5 m/s^2. At the same time, a cyclist rides South from the other end at a constant rate of 10.0 m/s. At what point on the bridge do they meet? How long does it take for them to meet?

$y = \Delta t = 36.357$ s or $\Delta t = -49.691$ s $x = \Delta d_{car} = 991.5$ m

Head Starts



In some situations, the objects might start moving at different times. In these cases, Δt_1 and Δt_2 are not equal.

If object 1 has the head start:	Let	$\Delta t_1 = y$ + head start	$\Delta t_2 = y$
If object 2 has the head start:	Let	$\Delta t_1 = y$	$\Delta t_2 = y$ + head start

To help you remember this, think in terms of the total time: the object that has the head start has more time to move. The object that starts later has less time to catch up. The problem can then be solved using the same procedures as above.

Example 2
A tortoise and a hare decide to run a 100.0 m race. The Hare is over confident and waits 30. s before they start to run, accelerating at a rate of 2.4 m/s^2. The tortoise accelerates at a rate of 0.75 m/s^2. At what distance will the hare catch up to the tortoise? Who wins the race?

$\Delta d_t = x$ $\Delta d_h = x$ $\Delta t_t = y + 30.$ s $\Delta t_h = y$ $y = -10.7571$ s or 38.0298 s $x = 1735.52$ m

Chase and Collision Problems

Applies to problems when objects are approaching each other, chasing each other or trying to get away from each other. For example, a police car chasing a speeding car, a person running after a bus, two trains heading toward each other.

Example 1: Diana and Ivan are in love and run towards each other from 75 m apart. Ivan starts from rest and accelerates at 1.08 m/s2 [N]. Diana runs at a constant velocity of 2.70 m/s [S]. How much time until they collide?

Example 2: Marisa is late for the bus so she is running east, as fast as she can (9.9 m/s). The bus pulls away, accelerating at 1.1 m/s2 [E] when Marisa 25 m west of the bus. When will Marisa catch the bus?

Even More Physics Questions

1. A dragster accelerates from rest at 14 m/s^2 [S]. During the acceleration, the dragster travels 450 m[S]. A parachute is then used to slow it down to a stop. The parachute causes the dragster to have an acceleration of 7.0 m/s^2 [N]. What is the total distance the dragster travelled from start to finish? **1400 m**

2. Superwoman is hovering 1.00 x 10^3 m above the ground when a person who is freely falling goes by at a terminal velocity of 140 km/h. Unfortunately for the person, their parachute does not open. Fortunately for the person, Superwoman is around. If it takes her 1.9 s to realize the person is in distress, what must her acceleration be if she is to catch the parachutist, just before he hits the ground 1.00 x 10^3 m below? NOTE: Terminal velocity is the maximum speed an object reaches when it is falling – terminal velocity can only occur if there is air resistance. Once an object reaches its terminal velocity, it will continue to fall at a constant speed. **3.5 m/s^2 [down]**

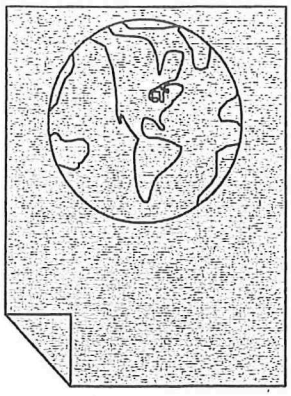
3. Two rugby players are separated by a distance of 37 m and are running directly towards each other. One is headed west and is accelerating from rest at 0.50 m/s^2. The other is moving at 3.1 m/s and is headed east.
   a. How much time will pass before they crash into each other? **7.5 s**
   b. How fast was the accelerating player going? **3.7 m/s**
   c. How far has each player run? **23 m from first player, 14 m from second player.**

4. A police car is stopped at a set of lights when a speeder passes him travelling south at 1.0 x 10^2 km/h. If the police car can accelerate at 3.6 m/ s^2,
   a. What is the minimum length of time that it would take to catch the speeder. Assume that the speeder does not change his/her velocity? **15 s**
   b. What is the minimum distance that the police care would have to go to catch the speeder? **430 m**
   c. What velocity would the police car have when it catches the speeder? **56 m/s [S]**

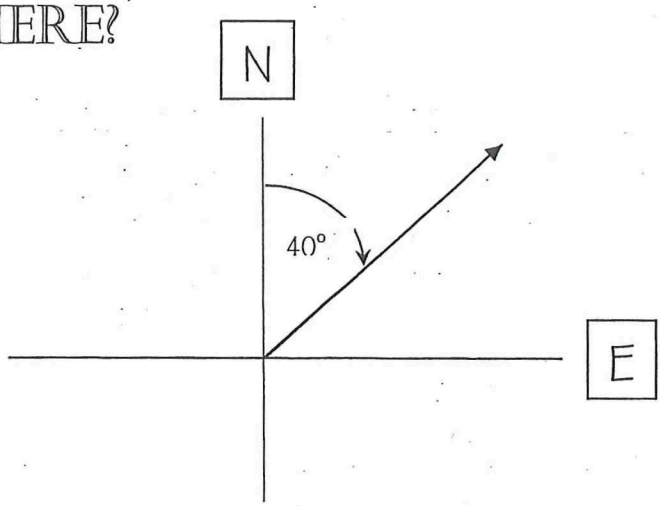
5. A trainer is desperately trying to catch Pikachu. At the instant that the trainer starts, Pikachu is 32 m east of the trainer and running east at 7.2 km/h, while accelerating at 0.40 m/s^2 [E]. The trainer jumps up and starts running, accelerating at 1.2 m/s^2 [E].
   a. How much time will elapse before the trainer catches up to Pikachu? **12 s**
   b. How far will the trainer have travelled during the chase? **83 m**
   c. What was Pikachu's average velocity during the chase? **4.4 m/s [E]**

6. Bulbasaur and Charmander have taken a dislike to each other. They have decided to settle their differences by running at each other from opposite ends of a field. By the time that concerned friends have turned up to try to avert the altercation, Bulbasuar is running north at 1.8 km/h and accelerating at 1.4 m/s^2 [N]. Charmander is running south at 2.7 km/h and accelerating at 1.0 m/s^2 [S]. They are separated by a distance of 40.0 m. What would be Bulbasaur's velocity when they reach each other? (express your answer in km/h). **28 km/h**

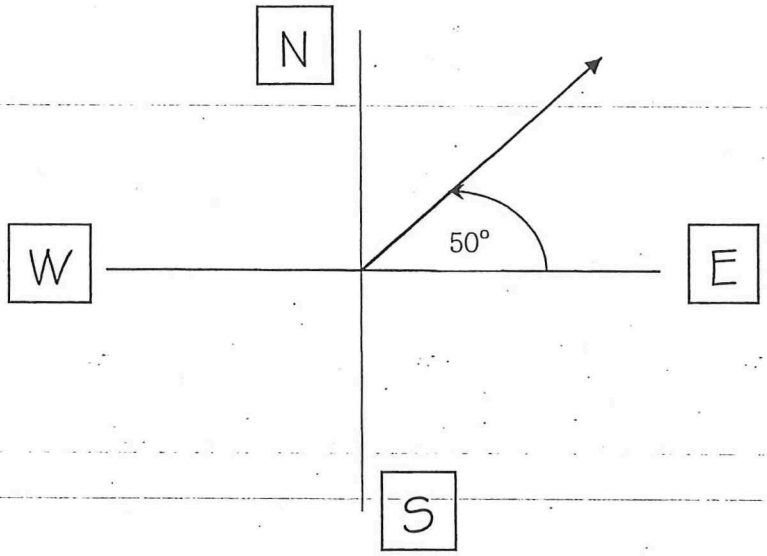
7. The Pink Ranger and the Red Ranger haven't seen each other for a long time. They are standing at opposite ends of a nice grassy field, separated by 115 m. The Red ranger notices the Pink Ranger and starts running west, towards the Pink Ranger, accelerating at 19440 km/h^2 [W]. The Pink Ranger is a little slow on the updates and does not notice the Red Ranger until he has been running for 3.00 s. At that instant, the Pink Ranger starts running towards the Red Ranger, accelerating at 0.90 m/s^2 [E]. How much time will pass, form when the Red Ranger started running before the two Rangers are reunited? **11 s**



IF YOU DON'T KNOW WHERE YOU ARE GOING ... HOW DO YOU EXPECT TO GET ANYWHERE?



go **North**	swing towards **East**	N 40 ° E	40 degrees East of North	40 ° E of N

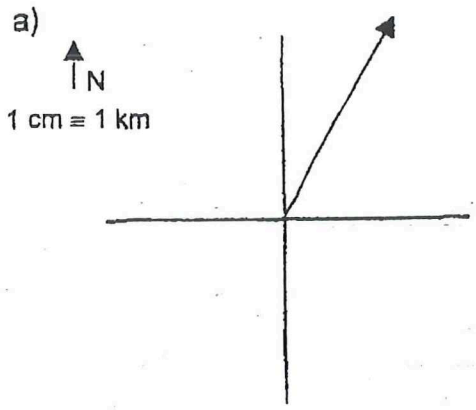


go **East**	swing towards **North**	E 50° N	50 degrees North of East	

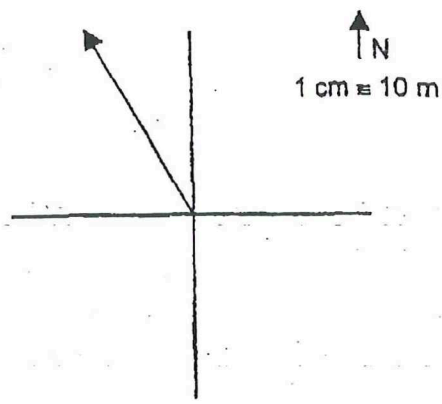
Drawing Vectors

1) What is the magnitude and direction of the following vectors?

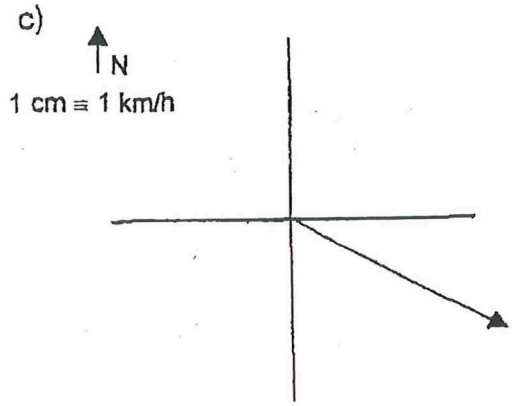
a)



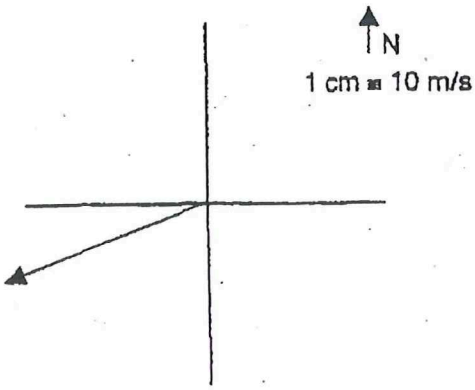
b)



c)



d)



2) Draw the following vectors to scale. Make sure you indicate the scale you used.

a) 3.0 m [N25°W]

b) 4.0 km/h [W50°S]

c) 3.5 km [E35°N]

d) 45 km [E15°S]

Vector Addition

The easiest way to learn how to add vectors is to look at it graphically.

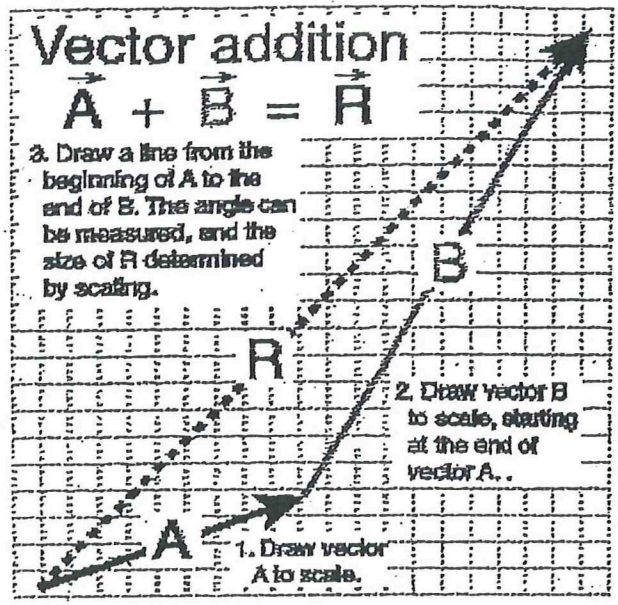
Vectors are added head-to-tail.

We can add any two vectors, $\vec{A}$ and $\vec{B}$, by placing the tail of $\vec{B}$ so that it meets the tip of $\vec{A}$.

The sum, $\vec{A} + \vec{B}$, is the vector from the tail of $\vec{A}$ to the tip of $\vec{B}$.

Note that you will get the same vector if you place the tip of $\vec{B}$ against the tail of $\vec{A}$. In other words, $\vec{A} + \vec{B}$ and $\vec{B} + \vec{A}$ are equal.

Graphical Vector Addition



Adding two vectors A and B graphically can be visualized like two successive walks, with the vector sum being the vector distance from the beginning to the end point. Representing the vectors by arrows drawn to scale, the beginning of vector B is placed at the end of vector A. The vector sum R can be drawn as the vector from the beginning to the end point.

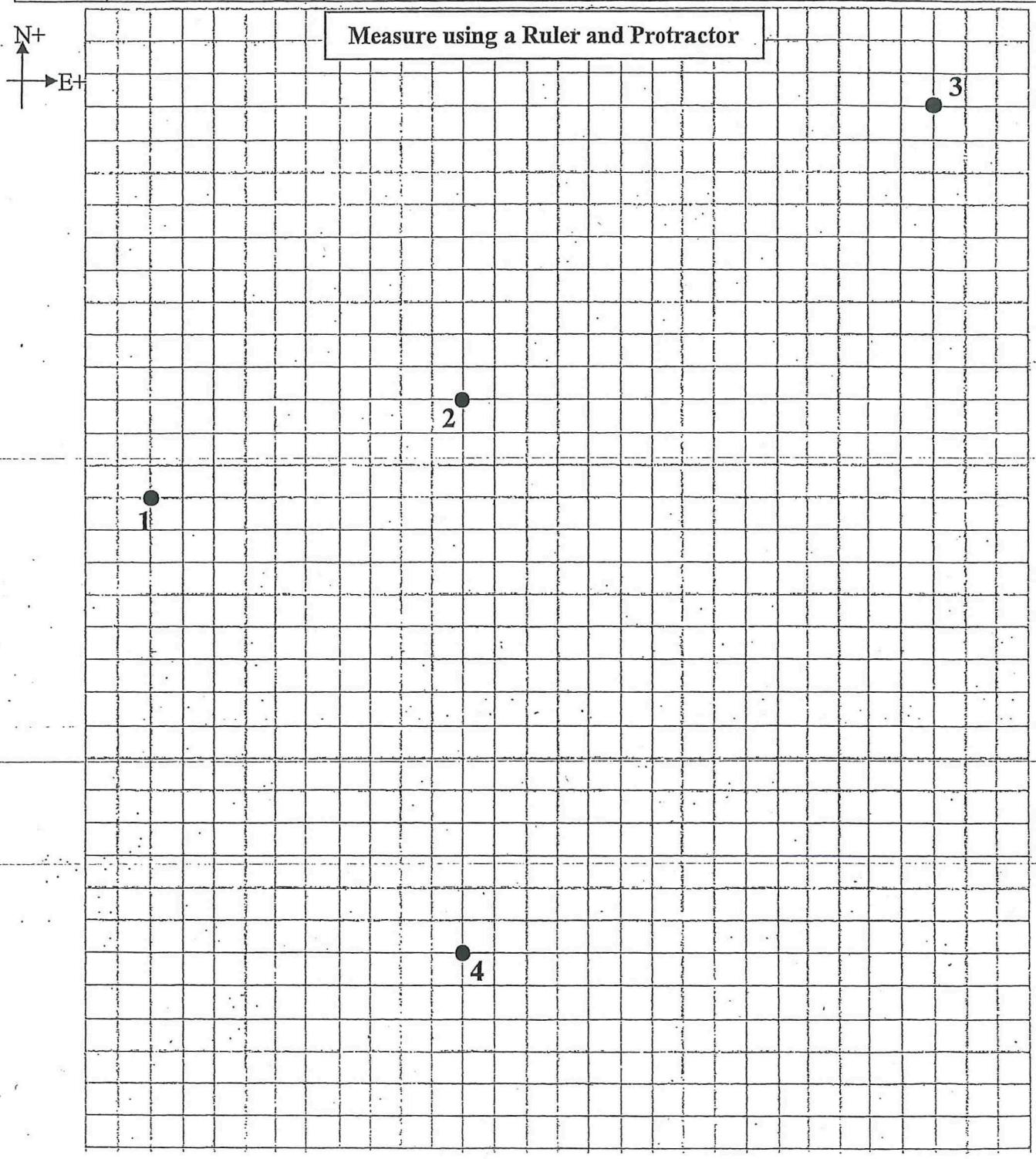
Text pg 75 # 2, 5, 6, 8, 9, 10, 12, 13, 15

1. A girl hikes 6.0 km[S55°W], 9.0 km[W], and 8.0 km[N]. The entire trip takes 5.0 h. Use the graphical method to answer these questions.
   (a) What is the total distance of the hiking trip?
   (b) What is the total displacement of her trip?
   (c) What is the average velocity of the trip?

Vector Addition Practice:

Create each of the following sets of vectors. Add the vectors graphically and determine the resultant (sum). Write the magnitude and direction of the resultant in the space next to each set.

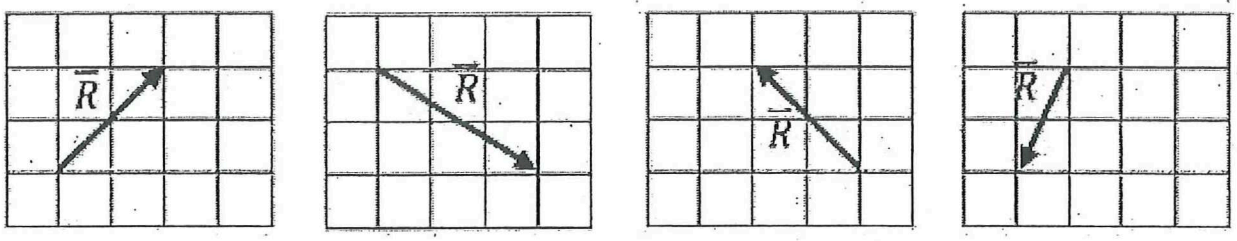
Trial	A	B	C	D	Resultant
1	7.0 cm, N	5.0 cm, E	None	None	
2	5.7 cm, 35° E of N	2.8 cm, 25° S of W	None	None	
3	3.1 cm, 15° W of S	5.7 cm, S	5.3cm, 30° E of N	None	
4	6.7 cm, 45° N of W	3.2 cm, 60° S of E	4.0cm, E	5.9 cm, 55°N of W	



The Vector Component Method of Adding/Subtracting Vectors

Part 1: Definition and Graphical Form of Vector Components

Draw $\vec{P}$ and $\vec{Q}$ only in vertical and horizontal axes where $\vec{R} = \vec{P} + \vec{Q}$



The drawn $\vec{P}$ and $\vec{Q}$ vectors are called **vector components** where **y-component** is $\vec{R}_y$ and **x-component** is $\vec{R}_x$

Every vector can be broken down into its h_____ (x) and v_____ (y) components.

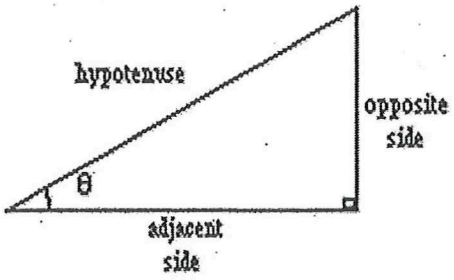
Every vector can be thought of as the hypotenuse (longest side) of a right triangle. The hypotenuse is the longest side of a right triangle

Trig Ratios:

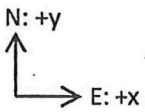
$$\text{SOH} \quad \sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\text{CAH} \quad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\text{TOA} \quad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$



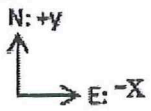
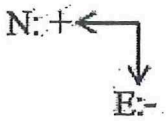
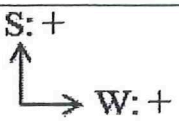
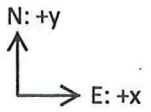
Given the following vector: $\vec{v}_i = 23\frac{m}{s}$ [S14°W] and the direction convention:



The x-component of $\vec{v}_i$ is: (leave 4 decimal places) _____

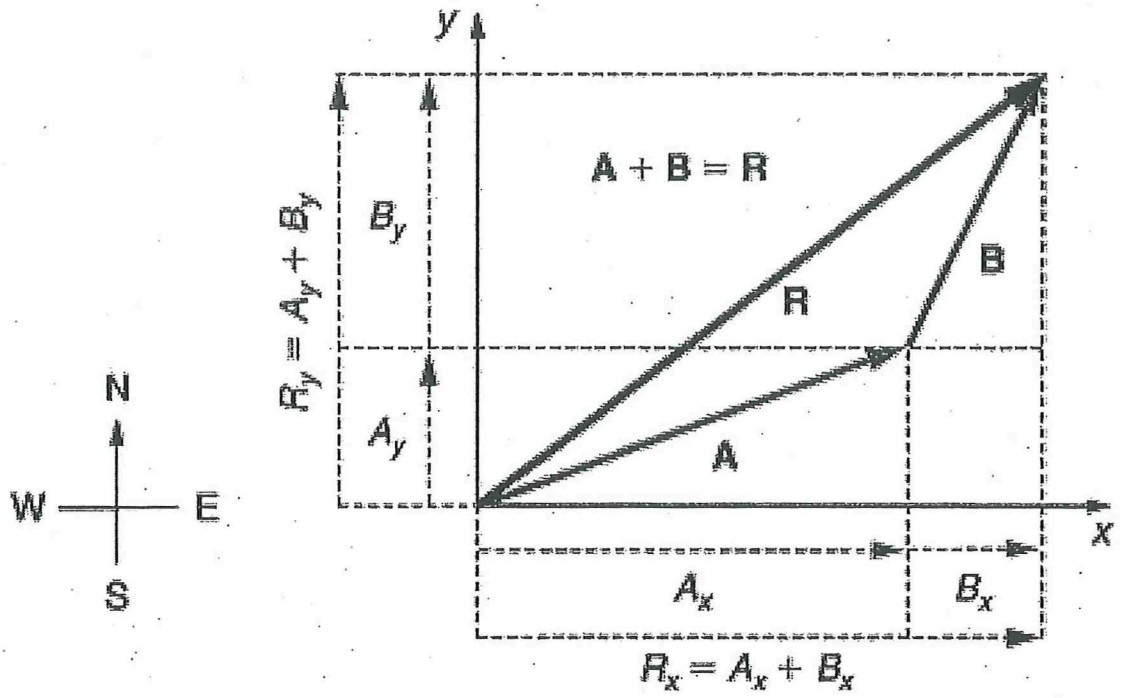
The y-component of $\vec{v}_i$ is: (leave 4 decimal places) _____

Given the following vectors and direction conventions, find the x- and the y-components. Leave answers with 4 decimal places.

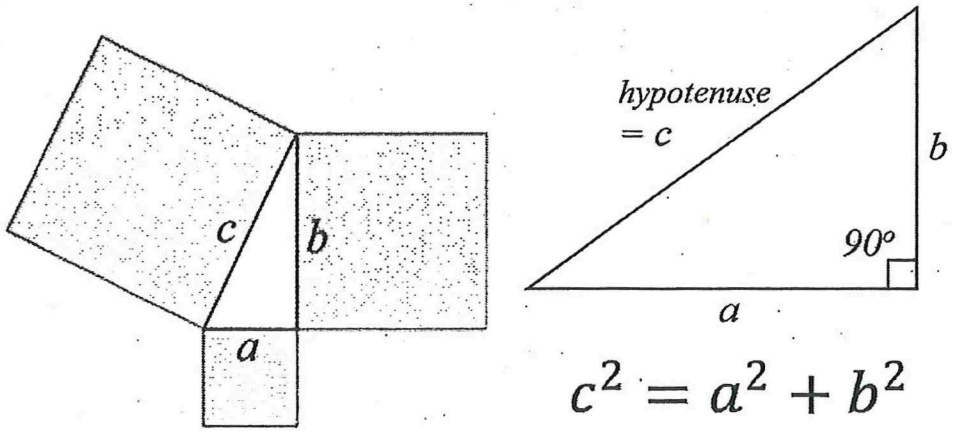
Vector (and sketch)	Direction Convention	x-component	y-component
$\vec{v}_i = 23\frac{m}{s}$ [S14°W]			
$\vec{v}_f = 16\frac{m}{s}$ [N14°W]			
$\overrightarrow{\Delta d} = 45\ km$ [S14°E]			
$\overrightarrow{\Delta d} = 45\ km$ [N14°E]			

How to Add Vectors:

1) Resolve each vector into its x and y components
2) Add all x components together. (This is the resultant vector's x-component, Rx)
3) Add all y components together. (This is the resultant vector's y-component, Ry)
4) Find the resultant vector by adding Rx to Ry. (Use Pythagorean theorem to get the magnitude. Use a trig ratio to get the direction.)



Pythagorean Theorem:



$$c^2 = a^2 + b^2$$

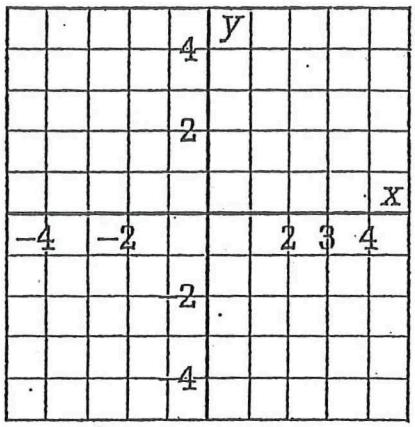
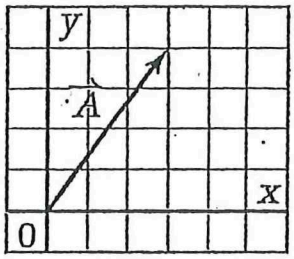
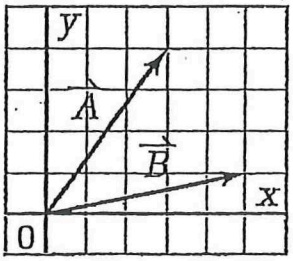
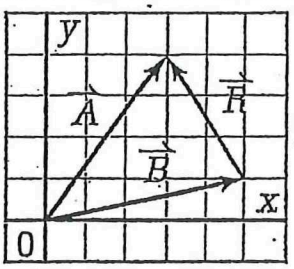
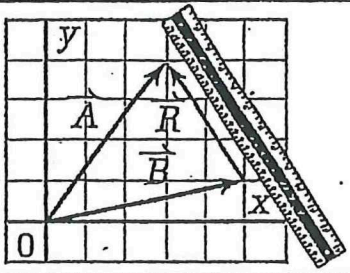
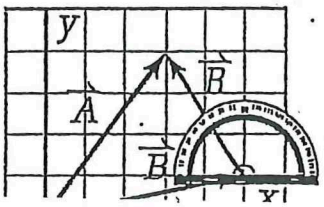
Vector Addition Practice:

Create each of the following sets of vectors. Add the vectors graphically and determine the resultant (sum). Write the magnitude and direction of the resultant in the space next to each set.

Trial	A	B	C	D	Resultant
1	7.0 cm, N	5.0 cm, E	None	None	
2	5.7 cm, 35° E of N	2.8 cm, 25° S of W	None	None	
3	3.1 cm, 15° W of S	5.7 cm, S	5.3cm, 30° E of N	None	
4	6.7 cm, 45° N of W	3.2 cm, 60° S of E	4.0cm, E	5.9 cm, 55° N of W	

When subtracting vectors, you have an option of two different methods. Table 3.3 describes the second procedure.

Table 3.3 Graphical Vector Subtraction ($\vec{R} = \vec{A} - \vec{B}$): Method Two

Procedural step	Graph
Establish a coordinate system and choose a scale.	
Place the first vector ($\vec{A}$) in the coordinate system.	
Place the tail of the second vector ($\vec{B}$) at the *tail* of the first vector ($\vec{A}$).	
Draw a vector from the *tip* of the second vector ($\vec{B}$) to the *tip* of the first vector ($\vec{A}$). Label this vector "$\vec{R}$."	
With a ruler, measure the length of $\vec{R}$.	
With a protractor, measure the angle between $\vec{R}$ and the horizontal axis.	

Vector Subtraction and Acceleration in Two Dimensions

Steps:

1. Write the equation *(e.g.* $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$*)*, rearrange to isolate the unknown vector, and enter values (magnitude and direction)
2. Change the sign of negative value to positive and change direction by 180° (e.g. north becomes south, [N30°E] becomes [S30°W])
3. Sketch a vector diagram to represent the new vector addition equation.
4. Solve. (Methods: sine/cosine laws, component method, scaled diagram)

Example 1: A school bus changes its velocity from 10 m/s [W] to 10 m/s [S60°W] in 5.0 s. Determine the following for the bus:

a. Change in velocity
b. Average acceleration during the turn

SPH3U Physics Kinematics

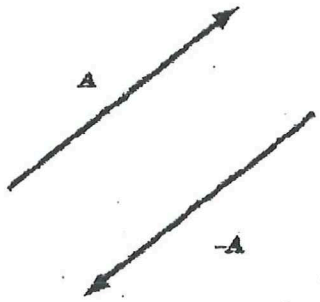
Vector Subtraction Problems

1. A bird travelling 3.0 m/s [N45°W] turns and heads south at the same speed. Determine the bird's change in velocity.
2. A car travelling at 15 m/s [N] executes a gradual turn, so that it then moves at 18m/s [E]. What is the car's change in velocity?
3. A boat's speed changes from 20 km/h [N] to 35 km/h s [N30°W] in 22 s. Find the boat's average acceleration during the turn.
4. After completing a turn, a cyclist is travelling east at 25 km/h. If her change in velocity was 15 km/h south, determine her initial velocity.

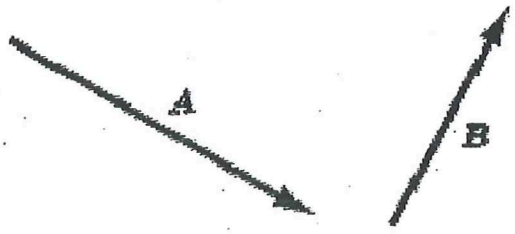
Answers:
1. 5.5m/s [S23°E], 2. ~~22m/s [W12°N]~~ 23m/s, 3. 0.91 km/h/s [W31°N], 4. 29 km/h [E31°S] N

Vector Subtraction

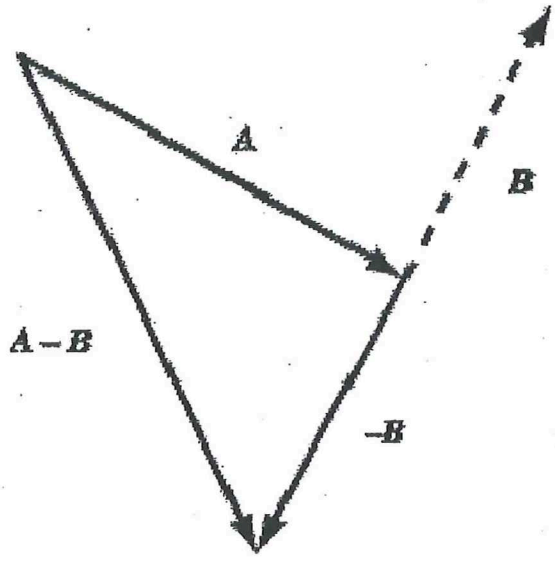
---You probably know that subtraction is the same thing as adding a negative: 8 – 5 is the same thing as 8 + (–5). The easiest way to think about vector subtraction is in terms of adding a negative vector. What's a negative vector? It's the same vector as its positive counterpart, only pointing in the opposite direction.



$A - B$, then, is the same thing as $A + (-B)$. For instance, let's take the two vectors A and B:



To subtract B from A, take a vector of the same magnitude as B, but pointing in the opposite direction, and add that vector to A, using either the tip-to-tail method or the parallelogram method.



Physics: Kinematics Review - Problem Solving

1. A bobsled has a constant acceleration of 2.0 m/s^2 [forwards], after having started from rest.
   a) What is its velocity, 5.0 s after it started?
   b) How far has it travelled in 5.0 s?
   c) What is its average velocity, over the first 5.0 s?
   d) How far has it travelled, by the time its speed reaches 40 m/s (2 s.d)?
   ANS: a) 1.0×10^1 m/s [F] b) 25 m c) 5.0 m/s [F] d) 4.0×10^2 m

2. While driving in a car, a woman sees an obstruction in the road. It takes 0.80 s for the woman to react and put her foot on the brake. Her car is travelling at 90.0 km/h [forwards].
   a) How far will the car travel before she puts her foot on the brake.
   b) Once the brake is applied, the car experiences a uniform acceleration of 9.3 m/s^2 [backwards]. What is the total displacement of the car, from the point where she sees the obstruction, to the point when she stops?
   ANS: a) 2.0×10^1 m b) 54 m [F]

3. At the instant when the traffic light turns green, an automobile starts with a constant acceleration of 1.8 m/s^2 [forwards]. At the same instant, a truck travelling forwards, with a constant speed of 8.5 m/s overtakes and passes the automobile.
   a) How far beyond the starting point will the automobile overtake the truck?
   b) What will the car's speed be, at this point?
   ANS: a) 8.0×10^1 m b) 17 m/s

4. An apple is thrown vertically upwards, with a speed of 18.7 m/s, from a window, in an apartment building. The apple hits the ground, 6.03 s later.
   a) How far, above the ground, is the window?
   b) With what velocity, does the apple hit the ground?
   c) How far above the window, does the apple reach?
   ANS: a) 65.6 m b) 40.5 m/s [down] c) 17.8 m

5. A car is driving east, at 36 km/h. The car accelerates uniformly to a velocity of 79.2 km/h [E], during which time it travels 80.0 m. The car maintains this speed for 10.0 s. Finally, the car comes to a stop, while accelerating uniformly at a rate of 2.75 m/s^2 [W]. What was the car's average velocity from the point at which it was driving at 36 km/h [E], to the point at which it is stopped?
   ANS: 17 m/s [E]

6. A stone is dropped into the water from a bridge, 44 m above the water. Another stone is thrown vertically downward 1.0 s after the first stone was dropped. Both stones strike the water at the same time. What was the initial velocity of the second stone?
   ANS: 12 m/s [down]

7. Two cars are 175 m apart, facing each other. At that instant, the first car is traveling at 9.0 km/h [W] and is accelerating at 3.6 m/s^2 [W]. The second car is traveling at a constant velocity of 21.5 m/s [E].
   a) How long will it take for the cars to hit each other?
   b) With what speed will the first car be travelling?
   ANS: a) 5.2 s b) 21 m/s

8. At a construction site, a bolt slips over the edge of a steel girder, which is 35 m above the ground. At the same instant an apple is thrown vertically upwards, from the ground, with a speed of 15 m/s.

a) When will the bolt and the apple be at the same height? (while they are both in the air?)

b) At what height, above the ground, will the two objects meet?

ANS: a) 2.3 s b) 8.3 m

Physics: Kinematics Review - Vector Diagrams/Components

1. A man walks 600 m [N43°E], then 500 m [N38°W], then 300 m [S61°W] and finally 400 m [S13°E]. Assuming that all measurements have 3 significant digits, find the man's resultant displacement.

ANS: 306 m [N13°W]

2. During a marathon, a person runs 3.57 km [W28°N], then turns and runs 4.61 km [S49°W] and finally runs 5.14 km [N12°E]. This takes them 1 hour and 12 minutes.

a) What was their average velocity during the run?

b) What was their average speed during the run?

ANS: a) 1.54 m/s [N57°W] b) 3.08 m/s

Not on test

3. A boat travelling at 3.0 m/s through the water, keeps its bow pointing north across a stream that flows west at 5.0 m/s. What is the velocity of the boat with respect to the shore?

ANS: 5.8 m/s [N59°W]

Not on test

4. An airplane maintains a heading due west at an air speed of 900 km/h. It is flying through a hurricane, with winds of 300 km/h, which are blowing southwest. What is the plane's velocity, relative to the ground? (Assume that all measurements have 3 s.d.)

ANS: 1130 km/h [S79°W]

Not on test

5. A plane has an airspeed of 470 km/h, with a heading of [N18°W]. An observer on the ground, notices that the plane is travelling at 420 km/h [N10°E]. What is the windspeed, and the direction that it is blowing?

ANS: 221 km/h [S81°E]

Not on test

6. A car, moving initially at 32 km/h [N], turns a corner and continues at 32 km/h [W]. The turn takes 3.0 s to complete.

a) What is the change in the car's velocity?

b) What is the average acceleration during the turn?

ANS: a) 45 km/h [SW] or [S45°W] b) 4.2 m/s^2 [SW] or [S45°W]

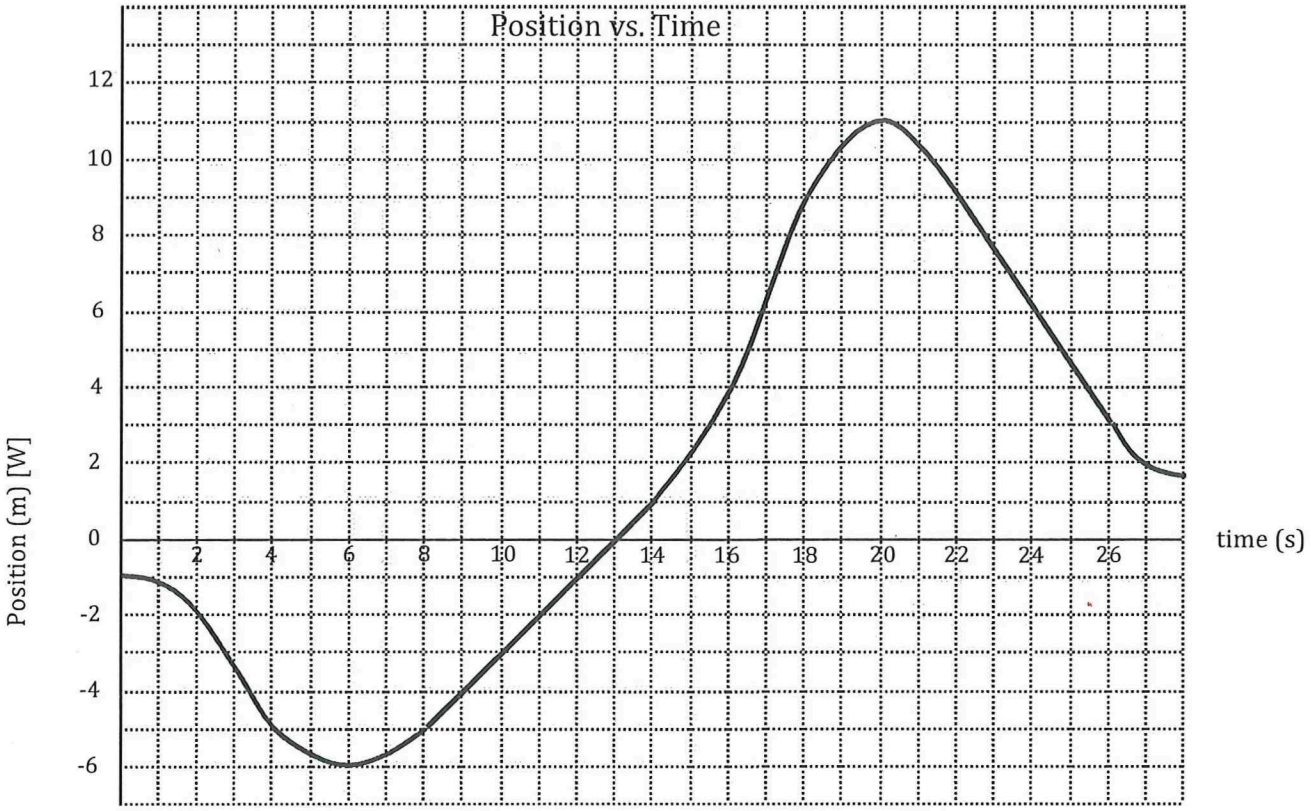
Not on test

7. A puck is sliding across the ice at 20 m/s [E] (2 s.d.). The puck is struck by a stick and moves at 30 m/s [S30°W] (2 s.d.). The puck is in contact with the hockey stick for 0.15 s. What is the puck's average acceleration, during its change in direction?

ANS: 290 m/s^2 [S53°W]

Physics: Kinematics Review – Graphing

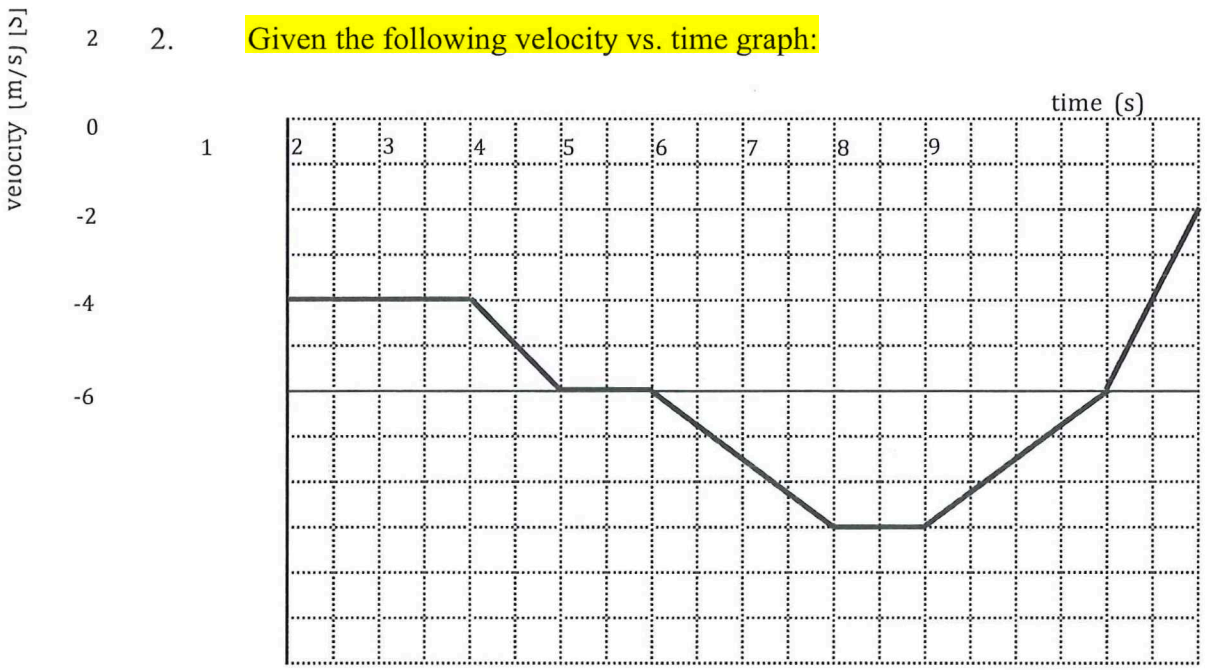
1. Given the following position vs. time graph:



a) At what time(s) was the object stopped?
b) At what time(s) did the object have an instantaneous velocity of 0 m/s?
c) At what time(s) did the object return to its starting position?
d) At what time(s) was the object at the reference position?
e) What was the object's average velocity during the following intervals?
   i) 0s to 6.0 s
   ii) 0 s to 18.0 s
   iii) 14.0 s to 22.0 s
   iv) the entire journey
f) What was the object's average speed during the same intervals as e?
g) What was the object's instantaneous velocity at:
   i) 2.0 s?
   ii) 16.0 s?
   iii) 27.0 s?
h) What was the object's instantaneous speed at the same points in time as g?
i) During what intervals was the object:
   i) west of the reference position?
   ii) east of the reference postion?
   iii) travelling west?
   iv) travelling east?
   v) travelling at a constant velocity?
   vi) speeding up?
   vii) slowing down?

Velocity vs. Time

2. Given the following velocity vs. time graph:



a) When was the object stopped?
b) When did the object have an instantaneous velocity of 0 m/s?
c) What was the object's instantaneous velocity at the following times?
   i) 1.0 s
   ii) 5.0 s
   iii) 9.5 s
d) What was the object's instantaneous speeds at the same times as in c?
e) During what intervals was the object:
   i) travelling south?
   ii) travelling north?
   iii) travelling at a constant velocity?
   iv) speeding up?
   v) slowing down?
f) Draw a displacement vs. time graph for the object.
g) Draw an acceleration vs. time graph for the object.
h) What was the object's average velocity during the following intervals?
   i) 0s to 6.0 s
   ii) 3.0 s to 9.0 s
   iii) the entire journey
i) What was the object's average speed during the same intervals as in h?

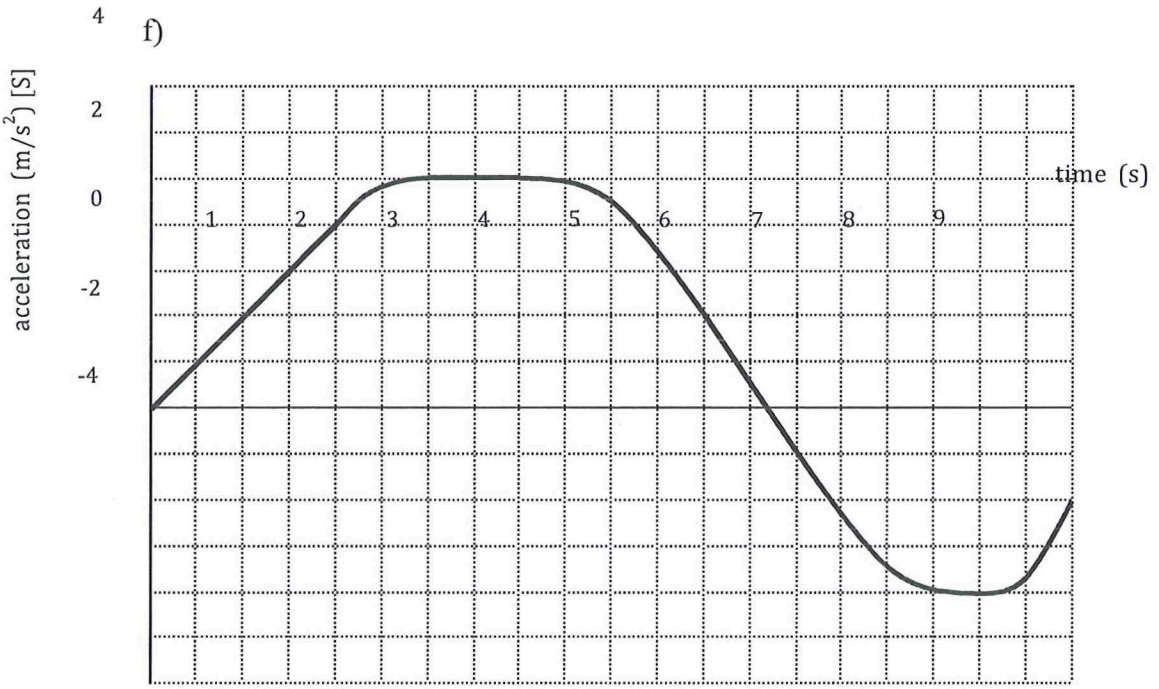
GRAPHING ANSWERS:

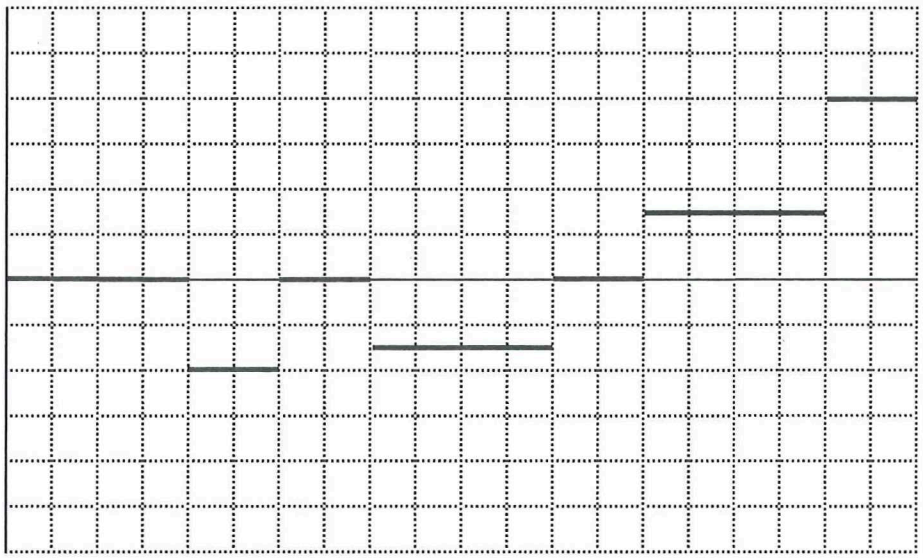
1. a) never
   b) 0s, 6.0 s, 20.0 s, 28.0 s
   c) 12.1 s
   d) 13.1 s
   e) i) 0.83 m/s [E]
      ii) 0.55 m/s [W]
      iii) 1.0 m/s [W]
      iv) 0.096 m/s [W]
   f) i) 0.83 m/s
      ii) 1.1 m/s
      iii) 1.5 m/s
      iv) 1.1 m/s
   g) i) 1.2 m/s [E]
      ii) 1.8 m/s [W]
      iii) 0.62 m/s [E]
   h) i) 1.2 m/s
      ii) 1.8 m/s
      iii) 0.62 m/s
   i) i) 13.1 s < t < 28.0 s
      ii) 0 s < t < 13.1 s
      iii) 6.0 s < t < 20.0 s
      iv) 0 s < t < 6.0 s, 20.0 s < t < 28.0 s
      v) 8.0s < t < 14.0 s, 22.0 s < t < 26.0 s
      vi) 0 s < t < 3.0 s, 6.0 s < t < 8.0 s, 14.0 s < t < 17.0 s, 20.0 s < t < 22.0 s
      vii) 3.0 s < t < 6.0 s, 17.0s < t < 20.0 s, 26.0 s < t < 28.0 s

2. a) 3.0 s < t < 4.0 s
   b) 9.0s (and at every instant between 3.0s and 4.0s)
   c) i) 2.0 m/s [S]
      ii) 1.5 m/s [N]
      iii) 2.0 m/s [S]
   d) i) 2.0 m/s
      ii) 1.5 m/s
      iii) 2.0 m/s
   e) i) 0 s < t < 3.0 s, 9.0 s < t < 10.0 s
      ii) 4.0 s < t < 9.0 s
      iii) 0 s < t < 2.0 s, 6.0 s < t < 7.0 s (and from 3.0 s < t < 4.0 s at a constant velocity of 0 m/s)
      iv) 4.0 s < t < 6.0 s, 9.0 s < t < 10.0 s
      v) 2.0 s < t < 3.0 s, 7.0 s < t < 9.0 s

Acceleration vs. Time

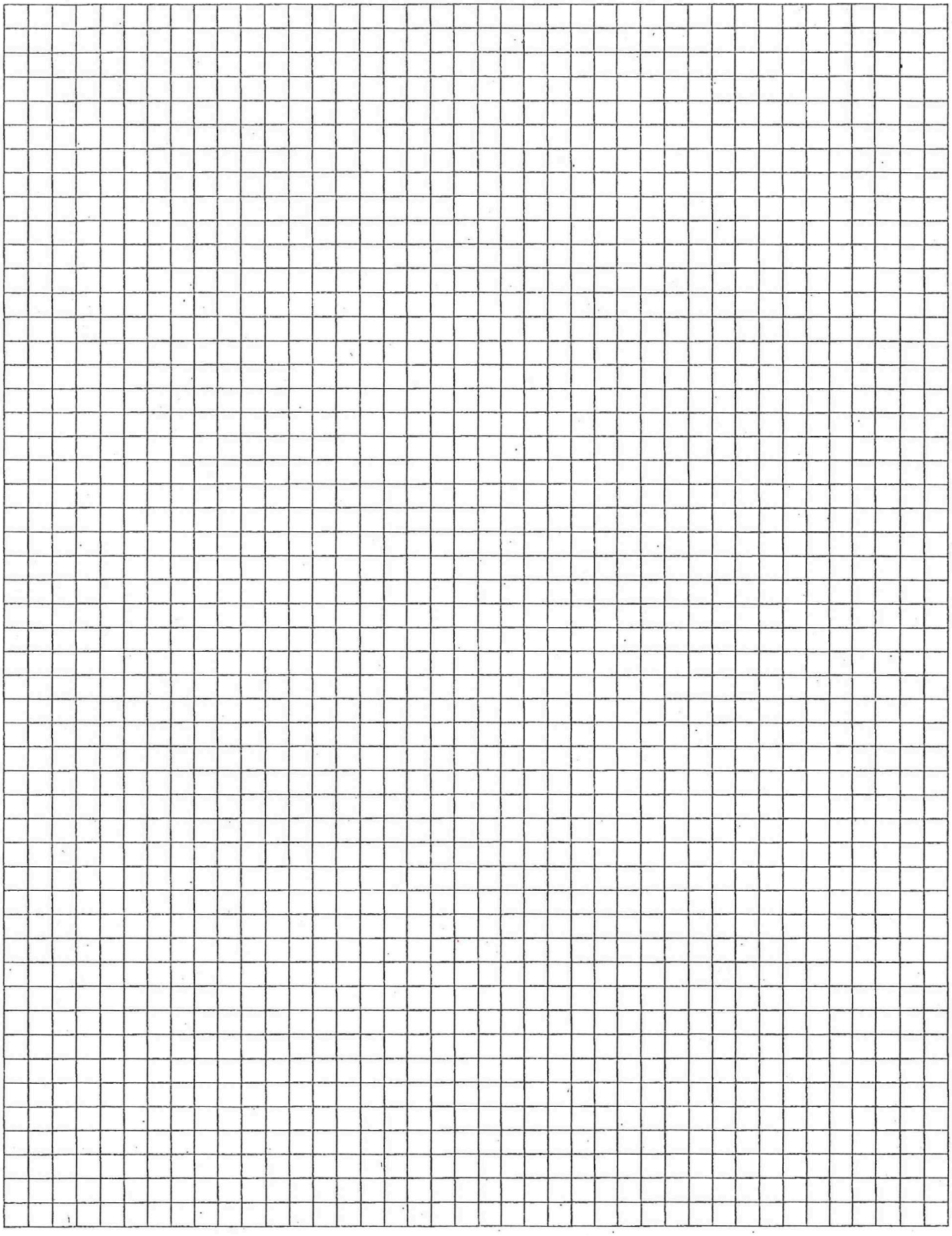
f)



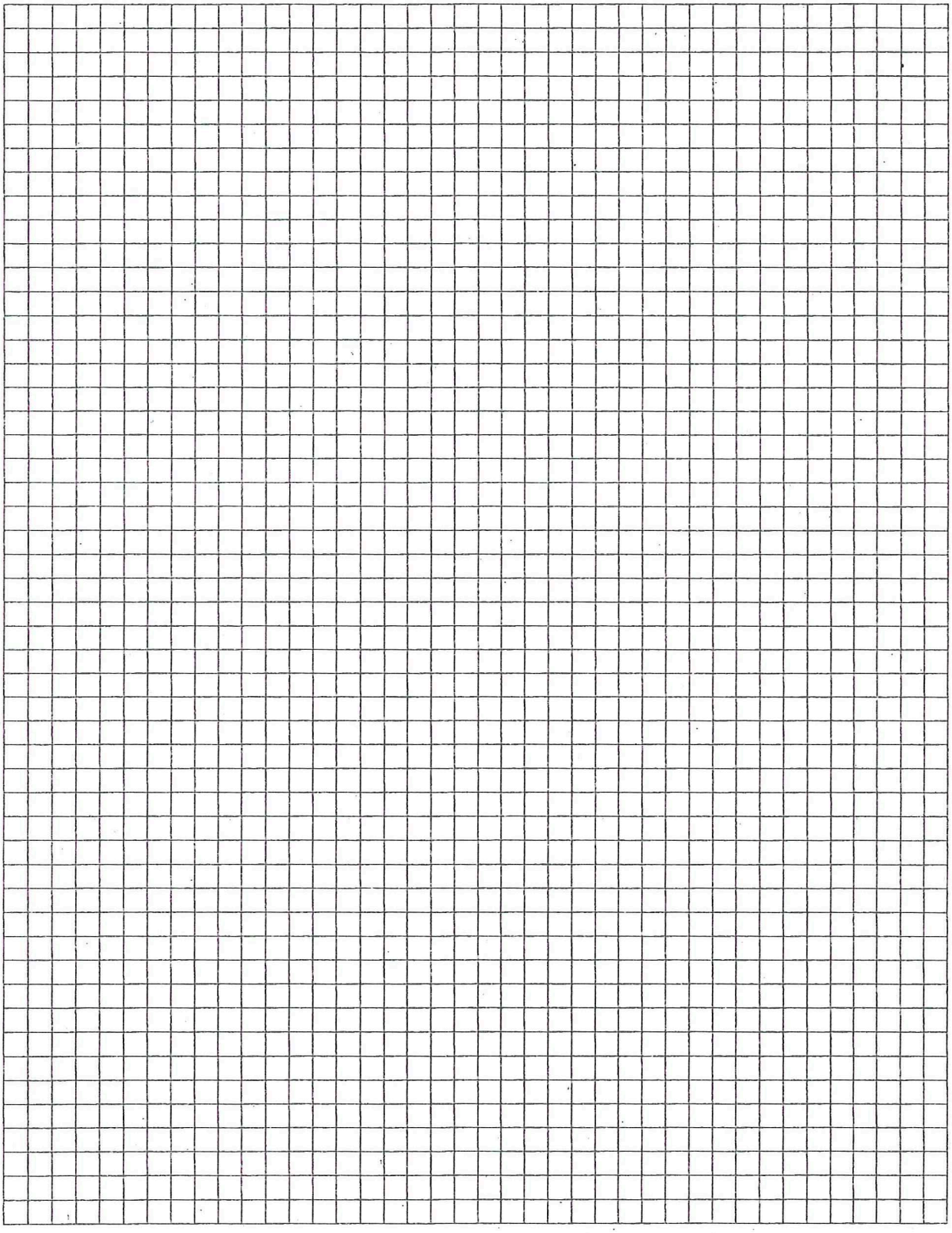


h) i) 0.33 m/s [S]
ii) 1.5 m/s [N]
iii) 0.2 m/s [N]

i) i) 1.3 m/s
ii) 1.5 m/s
iii) 1.6 m/s



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